Youssef Mohamed Ibrahim

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EDUCATION

Alamein International University *Alamein, Egypt | Sept 2022 – Present*

B.Sc. in Computer Engineering — Artificial Intelligence Program *Cumulative GPA: 3.68 / 4.0*

Saint-Marc College *Alexandria, Egypt | Sept 2009 – July 2022*

SKILLS

Programming Languages: Python, Java, JavaScript, SQL, R, HTML/CSS, Verilog, Embedded C, Bash

Languages: Arabic: Native English: Fluent `French: Fluent

Frameworks & Tools: React.js, Node.js, Docker, Kubernetes, Terraform, Kafka, TensorFlow, Keras, scikit-learn, OpenCV, Qiskit, PyTorch, Git&Github

Domains: Machine Learning, NLP, Computer Vision, Quantum Computing, Deep Learning, Full-Stack Web

WORK EXPERIENCE & CERTIFICATES

Quantum Computing Trainee — QWorld & Alamein International University *Jun 2025 – Sept 2025*

- Completed the 12-week QCourse110-1 program covering quantum measurement, single-qubit and controlled gates, circuit design, teleportation and superdense coding protocols, and quantum algorithms including Deutsch–Jozsa and Grover's search.
- Finished with a grade of 294/300 and delivered an individual final project implementing an Enhanced Grover's Algorithm with adaptive phase rotation in Qiskit.

Technical Support Agent *2023*

- Guided customers through connectivity, speed, and device setup issues over phone and chat, diagnosing problems using internal network tools and escalating complex cases when needed.

Certificates → Click Here

PROJECTS

SENTINEL+ — Hybrid AI-Generated Text Detector | *Python, PyTorch, RoBERTa, scikit-learn → Here*

- Built a heterogeneous ensemble fusing RoBERTa semantic embeddings (768-d), TF-IDF n-grams (10,000-d), and a statistical perplexity feature through a soft-voting classifier of linear and tree-based models.
- Reached 0.9960 accuracy and 0.9992 AUC-ROC on a 20,000-sample dataset covering GPT-4, GPT-4o, LLaMA-2, and Mistral outputs; used LIME for prediction-level interpretability.

MediCare Clinic — Cloud-Deployed Appointment System | *React, Node.js, Express, SQLite → Here*

- Designed a full-stack SPA with role-based access for patients and doctors, JWT authentication, bcrypt password hashing, and a normalized three-table relational schema for doctors, patients, and appointments.

Smart Classroom Assistant | *Python, OpenCV, scikit-learn → Here*

- Developed a multi-modal classroom tool that takes attendance through face recognition, estimates mood with a pre-trained CNN (happy, tired, sad), transcribes student questions via speech-to-text, and classifies them into subject topics using a TF-IDF model.

Intelligent E-Commerce Recommender | *Python, scikit-learn, Flask → Here*

- Implemented three recommendation approaches — collaborative filtering with four similarity and matrix-factorization variants, content-based, and knowledge-based — plus an evaluation module comparing them across four metrics and generating per-item explanations.

Enhanced Grover's Search Algorithm | *Python, Qiskit → Here*

- Implemented both the standard and an enhanced variant of Grover's search that applies adaptive phase rotation to reduce the iteration count on unstructured search problems; benchmarked circuits on Qiskit Aer and analyzed success probability across oracle sizes.

Trust Guard | *Python, React → Here*

- a full-stack AI platform built with React Native, FastAPI, Playwright, and SerpAPI to combat deceptive marketing and greenwashing in Egyptian e-commerce, utilizing strictly-grounded Grok/Llama LLMs to analyze real-time data, calculate deterministic credibility scores, verify product claims, and recommend trusted alternatives.